Module 8: Cellular Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. What is the main product of mitosis?
    - A. Four haploid gametes
    - B. Two genetically identical daughter cells
    - C. A single haploid cell
    - D. Cells with randomly scrambled DNA
    - Answer: B
    - Why: Mitosis produces two genetically identical daughter cells for growth and repair. Meiosis makes four haploid gametes.
2. Meiosis reduces the chromosome number by using:
    - A. two rounds of DNA copying and no division
    - B. no DNA replication at all
    - C. extra chromosomes added each cycle
    - D. one round of DNA copying followed by two divisions
    - Answer: D
    - Why: One replication followed by two divisions halves the chromosome number, producing haploid gametes.
3. Crossing over and independent assortment both act to:
    - A. increase genetic variation among gametes
    - B. produce identical gametes every time
    - C. occur only during mitosis
    - D. remove all differences between offspring
    - Answer: A
    - Why: Both processes shuffle alleles during meiosis, generating genetic variation among offspring.

4. If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:
    - A. exactly the normal number every time
    - B. no DNA whatsoever
    - C. an extra or a missing chromosome
    - D. twice as many cells
    - Answer: C
    - Why: Nondisjunction gives a gamete an abnormal chromosome count, which can lead to conditions such as trisomy.